Chapter 7. Complex Measures

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Overview

So far we have considered positive measures and integrals with respect to them. This chapter allows measures to take negative or complex values. If a positive measure is given on a σ-algebra, every other measure is the sum of a part expressible as an integral with respect to that measure and a part that is not (singular) — this is the Lebesgue–Radon–Nikodym theorem, the core of measure theory. Applying it to Lebesgue measure restricted to the Borel sets, we find all Borel measures on the real line.

7.1 Signed Measures and Complex Measures

7.1.1 Definitions

A function $\lambda : \mathcal{S} \to [0, \infty]$ on a σ-algebra $\mathcal{S}$ of a set X satisfying

$$\{A_n \in \mathcal{S} : n = 1, 2, \dots\} \text{ pairwise disjoint } \implies \lambda\Big(\bigcup_{n=1}^{\infty} A_n\Big) = \sum_{n=1}^{\infty} \lambda(A_n) \tag{1}$$

with $\lambda(\emptyset) = 0$ was called a positive measure.

Definition (signed measure, complex measure)

A function $\lambda : \mathcal{S} \to \mathbb{R}^*$ satisfying (1) and $\lambda(\emptyset) = 0$ is a **signed measure**. A function $\lambda : \mathcal{S} \to \mathbb{C}$ satisfying (1) is a **complex measure**.

In both definitions the convergence of $\sum \lambda(A_n)$ is part of the assumption; for signed measures, convergence to $+\infty$ or $-\infty$ is allowed. If $E, F \in \mathcal{S}$ with $\lambda(E) = \infty$ and $\lambda(F) = -\infty$, then in

$$\lambda(E \cup F) = \lambda(E \setminus F) + \lambda(F \setminus E) + \lambda(E \cap F)$$

both ∞ and $-\infty$ would appear on the right. Hence *a signed measure cannot take both values* $\pm\infty$. Sums and scalar multiples of complex measures are complex measures.

Example 1

For a measurable function h on $\mathbb{R}$ define

$$\lambda(E) = \int_E h(x)\, dx, \qquad E \in \mathcal{M}. \tag{2}$$

If $h \geq 0$, λ is a positive measure by Theorem 1.2.6. If $h : \mathbb{R} \to \mathbb{R}^*$ is measurable and at least one of h_+, h_- is integrable, λ defined by (2) is a signed measure. If $h : \mathbb{R} \to \mathbb{C}$ is integrable, λ is a complex measure.

Given a complex measure $\lambda : \mathcal{S} \to \mathbb{C}$, let $\lambda_1(E), \lambda_2(E)$ be the real and imaginary parts of $\lambda(E)$. Then λ_1, λ_2 are signed measures and

$$\lambda = \lambda_1 + i\lambda_2. \tag{3}$$

So we first study signed measures, showing that a signed measure is a difference of positive measures — just as a real function is $f = f_+ - f_-$.

7.1.2 The Hahn decomposition

Definition (positive and negative sets)

Let λ be a signed measure on a σ-algebra $\mathcal{S}$ of X. A set $P \in \mathcal{S}$ is **positive** if

$$E \in \mathcal{S},\ E \subset P \implies \lambda(E) \geq 0,$$

and **negative** if

$$E \in \mathcal{S},\ E \subset P \implies \lambda(E) \leq 0.$$

For example, for the signed measure of Example 1, $\{x : h(x) \geq 0\}$ and $\{x : h(x) < 0\}$ are positive and negative sets. A measurable subset of a positive set is positive. If $\{P_n : n = 1, 2, \dots\}$ are positive sets, their union P is positive: for $E \subset P$, $E \in \mathcal{S}$, put

$$E_n = (E \cap P_n) \setminus (P_1 \cup \cdots \cup P_{n-1}), \qquad n = 1, 2, \dots \quad (P_0 = \emptyset);$$

then $E_n \subset P_n$, so $\lambda(E_n) \geq 0$, and $\lambda(E) = \sum_n \lambda(E_n) \geq 0$.

Theorem 7.1.1 (Hahn decomposition)

If λ is a signed measure on a σ-algebra $(X, \mathcal{S})$, then X splits into a positive set and a negative set. If (P_1, Q_1) and (P_2, Q_2) are two such splittings, then for every $E \in \mathcal{S}$

$$\lambda(E \cap P_1) = \lambda(E \cap P_2), \qquad \lambda(E \cap Q_1) = \lambda(E \cap Q_2).$$

Proof. First assume λ does not take the value $+\infty$. Define

$$\alpha = \sup\{\lambda(P) : P \text{ is a positive set}\};$$

$\emptyset$ is positive, so $\alpha \geq 0$. Choose positive sets $\langle P_n \rangle$ with $\alpha = \lim_n \lambda(P_n)$; their union P is positive. For each n, $P \setminus P_n \subset P$ is positive, so $\lambda(P \setminus P_n) \geq 0$ and

$$\lambda(P_n) \leq \lambda(P_n) + \lambda(P \setminus P_n) = \lambda(P) \leq \alpha.$$

Hence $\lambda(P) = \alpha < \infty$. Now put $Q = X \setminus P$. If a measurable $E \subset Q$ is positive, then $E \cup P$ is positive, so

$$\alpha \geq \lambda(E \cup P) = \lambda(E) + \lambda(P) = \lambda(E) + \alpha \geq \alpha,$$

and therefore $\lambda(E) = 0$. That is,

$$E \in \mathcal{S},\ E \subset Q,\ E \text{ positive} \implies \lambda(E) = 0. \tag{4}$$

It remains to show Q is negative. Take any $E \in \mathcal{S}$ with $E \subset Q$. If E is positive, then $\lambda(E) = 0$ by (4) and we are done. If E is not positive, it has subsets of negative measure; let n_1 be the smallest natural number for which there is $F_1 \subset E$ with $\lambda(F_1) < -\frac{1}{n_1}$, and choose such F_1. If $E \setminus F_1$ is positive, then $\lambda(E \setminus F_1) = 0$ by (4), so $\lambda(E) = \lambda(F_1) < 0$ and we are done. If $E \setminus F_1$ is not positive, repeat: if $E \setminus (F_1 \cup \cdots \cup F_{k-1})$ is positive, its measure is 0 and $\lambda(E) = \sum_{i=1}^{k-1} \lambda(F_i) < 0$; otherwise choose the smallest natural number n_k and $F_k \in \mathcal{S}$ with

$$F_k \subset E \setminus (F_1 \cup \cdots \cup F_{k-1}), \qquad \lambda(F_k) < -\frac{1}{n_k}.$$

If the process never stops, put

$$F = E \setminus \Big[\bigsqcup_{k=1}^{\infty} F_k \Big].$$

Clearly $\langle n_k \rangle$ is nondecreasing.

If $\langle n_k \rangle$ is bounded, then $\lambda(\bigsqcup_k F_k) = -\infty$, and since $\lambda(F) < \infty$, $\lambda(E) = \lambda(\bigsqcup_k F_k) + \lambda(F) = -\infty$, and we are done.

If $\langle n_k \rangle$ is unbounded, we show F is positive. Suppose F contains $G \in \mathcal{S}$ with $\lambda(G) < 0$. Since $n_k \to \infty$, we can choose k with $\frac{1}{n_k - 1} < -\lambda(G)$. Then

$$G \subset F \subset E \setminus (F_1 \cup \cdots \cup F_{k-1}), \qquad \lambda(G) < -\frac{1}{n_k - 1},$$

contradicting the minimality in the choice of n_k. So F is positive, $\lambda(F) = 0$ by (4), and $\lambda(E) = \sum_{k=1}^{\infty} \lambda(F_k) < 0$. Hence Q is negative.

The case where λ does not take $-\infty$ is proved in the same way (apply the above to $-\lambda$).

For the second part, see Exercise 7.1.1. ■

Exercise 7.1.1

Prove the second part of the Hahn decomposition theorem.

Solution. Let $E \in \mathcal{S}$. The set $E \cap (P_1 \setminus P_2) = E \cap P_1 \cap Q_2$ is a subset of the positive set P_1 and of the negative set Q_2, so its measure is both ≥ 0 and ≤ 0, i.e. 0. Likewise $\lambda(E \cap (P_2 \setminus P_1)) = 0$. Hence

$$\lambda(E \cap P_1) = \lambda(E \cap P_1 \cap P_2) + \lambda(E \cap P_1 \cap Q_2) = \lambda(E \cap P_1 \cap P_2) = \lambda(E \cap P_2).$$

The same argument with Q_1, Q_2 gives $\lambda(E \cap Q_1) = \lambda(E \cap Q_2)$. □

7.1.3 Jordan decomposition and total variation

Given a Hahn decomposition $X = P \sqcup Q$ for λ, define for $E \in \mathcal{S}$

$$\lambda_+(E) = \lambda(E \cap P), \qquad \lambda_-(E) = -\lambda(E \cap Q).$$

By Theorem 7.1.1 these do not depend on the decomposition. λ_+ and λ_- are positive measures and

$$\lambda(E) = \lambda_+(E) - \lambda_-(E), \qquad E \in \mathcal{S}.$$

Also define

$$|\lambda|(E) = \lambda_+(E) + \lambda_-(E);$$

$|\lambda|$ is a positive measure as well.

Exercise 7.1.2

If λ is defined by (2), show that for every $E \in \mathcal{M}$

$$\lambda_+(E) = \int_E h_+, \qquad \lambda_-(E) = \int_E h_-, \qquad |\lambda|(E) = \int_E |h|.$$

Solution. $P = \{h \geq 0\}$, $Q = \{h < 0\}$ is a Hahn decomposition: for $E \subset P$, $\lambda(E) = \int_E h \geq 0$, and for $E \subset Q$, $\lambda(E) \leq 0$. Since $h\chi_P = h_+$ and $h\chi_Q = -h_-$,

$$\lambda_+(E) = \int_{E \cap P} h = \int_E h_+, \qquad \lambda_-(E) = -\int_{E \cap Q} h = \int_E h_-,$$

and $|\lambda|(E) = \int_E (h_+ + h_-) = \int_E |h|$. □

Proposition 7.1.2

If λ is a signed measure on a σ-algebra $\mathcal{S}$ of X, then for every $E \in \mathcal{S}$

$$|\lambda|(E) = \sup\Big\{ \sum_{n=1}^{\infty} |\lambda(E_n)| : \ E_n \in \mathcal{S}, \ E = \bigsqcup_{n=1}^{\infty} E_n \Big\}. \tag{5}$$

Proof. Let α be the right side of (5). If (P, Q) is a Hahn decomposition, then

$$|\lambda|(E) = \lambda_+(E) + \lambda_-(E) = |\lambda(E \cap P)| + |\lambda(E \cap Q)| \le \alpha.$$

Conversely, if $E_n \in \mathcal{S}$ and $E = \bigsqcup_{n=1}^{\infty} E_n$, then

$$\sum_{n=1}^{\infty} |\lambda(E_n)| = \sum_{n=1}^{\infty} |\lambda_+(E_n) - \lambda_-(E_n)| \le \sum_{n=1}^{\infty} \left(\lambda_+(E_n) + \lambda_-(E_n)\right) = \sum_{n=1}^{\infty} |\lambda|(E_n) = |\lambda|(E),$$

so $\alpha \le |\lambda|(E)$. ∎

7.1.4 Total variation of a complex measure

From now on let λ be a complex measure on a σ-algebra $(X, \mathcal{S})$ and define $|\lambda|(E)$ for $E \in \mathcal{S}$ by (5). Clearly $|\lambda(E)| \le |\lambda|(E)$ for every $E \in \mathcal{S}$.

Proposition 7.1.3

If λ is a complex measure on a σ-algebra $\mathcal{S}$ of X, then $E \mapsto |\lambda|(E)$ is a positive measure.

Proof. For a disjoint sequence $\langle E_n \rangle$ in $\mathcal{S}$ with union E we must show

$$|\lambda|(E) = \sum_{n=1}^{\infty} |\lambda|(E_n).$$

If $E = \bigsqcup_{m=1}^{\infty} F_m$, then for all m, n

$$F_m = \bigsqcup_{n=1}^{\infty} (E_n \cap F_m), \qquad E_n = \bigsqcup_{m=1}^{\infty} (E_n \cap F_m).$$

Hence

$$\begin{aligned} \sum_{m=1}^{\infty} |\lambda(F_m)| &= \sum_{m=1}^{\infty} \Big| \sum_{n=1}^{\infty} \lambda(E_n \cap F_m) \Big| \le \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} |\lambda(E_n \cap F_m)| \\ &= \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} |\lambda(E_n \cap F_m)| \le \sum_{n=1}^{\infty} |\lambda|(E_n), \end{aligned}$$

and taking the supremum over partitions $\{F_m\}$ gives $|\lambda|(E) \le \sum_n |\lambda|(E_n)$. For the reverse, let $\epsilon > 0$ and choose for each E_n a partition $\{E_{n,m} : m \in \mathbb{N}\}$ with

$$|\lambda|(E_n) < \sum_{m=1}^{\infty} |\lambda(E_{n,m})| + \frac{\epsilon}{2^n}, \qquad n = 1, 2, \ldots$$

Then $\{E_{n,m} : m, n = 1, 2, \ldots\}$ is a partition of E, so

$$\sum_{n=1}^{\infty} |\lambda|(E_n) \le \sum_{m,n=1}^{\infty} |\lambda(E_{n,m})| + \epsilon \le |\lambda|(E) + \epsilon.$$

■

By (3), a complex measure λ is a linear combination of positive measures:

$$\lambda = (\lambda_{1+} - \lambda_{1-}) + i(\lambda_{2+} - \lambda_{2-}). \tag{6}$$

If $\{E_n\}$ is a partition of $E \in \mathcal{S}$, then

$$\sum_{n=1}^{\infty} |\lambda(E_n)| \le \sum_{n=1}^{\infty} \big(\lambda_{1+}(E_n)+\lambda_{1-}(E_n)+\lambda_{2+}(E_n)+\lambda_{2-}(E_n)\big) = \lambda_{1+}(E)+\lambda_{1-}(E)+\lambda_{2+}(E)+\lambda_{2-}(E),$$

so for every $E \in \mathcal{S}$

$$|\lambda|(E) \le \lambda_{1+}(E) + \lambda_{1-}(E) + \lambda_{2+}(E) + \lambda_{2-}(E).$$

Since λ is a complex measure, the values on the right are always finite. Hence $|\lambda|(E) < \infty$ for every $E \in \mathcal{S}$; in particular $|\lambda|(X) < \infty$, and we define

$$\|\lambda\| = |\lambda|(X).$$

Note that positive measures are in general not complex measures. A positive measure with $\mu(X) < \infty$ is a **finite positive measure**; finite positive measures are of course complex measures.

Exercise 7.1.3

For a σ-algebra $\mathcal{S}$ on X, show that the set of all complex measures on $\mathcal{S}$ is a vector space and that $\lambda \mapsto \|\lambda\|$ is a norm on it.

Solution. If λ, ρ are complex measures and $a \in \mathbb{C}$, then $(\lambda+\rho)(\emptyset) = 0$ and for disjoint A_n,

$$(\lambda+\rho)\Big(\bigsqcup A_n\Big) = \sum \lambda(A_n) + \sum \rho(A_n) = \sum (\lambda+\rho)(A_n)$$

(sum of convergent series); similarly for $a\lambda$. So they form a vector space. Norm: $\|\lambda\| \ge 0$. If $\|\lambda\| = 0$ then $|\lambda(E)| \le |\lambda|(E) \le |\lambda|(X) = 0$ for all E, so $\lambda = 0$. For any partition $\{E_n\}$ of X, $\sum |a\lambda(E_n)| = |a| \sum |\lambda(E_n)|$, so $|a\lambda| = |a||\lambda|$ and $\|a\lambda\| = |a|\|\lambda\|$. Finally $\sum |(\lambda+\rho)(E_n)| \le \sum |\lambda(E_n)| + \sum |\rho(E_n)| \le \|\lambda\| + \|\rho\|$; taking the supremum gives $\|\lambda+\rho\| \le \|\lambda\| + \|\rho\|$. □

7.2 The Radon–Nikodym Theorem and the Lebesgue Decomposition

7.2.1 Absolute continuity and singularity

Definition

Let μ be a positive measure and λ a signed or complex measure on a σ-algebra $\mathcal{S}$ of X.

- λ is **absolutely continuous** with respect to μ, written $\lambda \ll \mu$, if

$$E \in \mathcal{S},\ \mu(E) = 0 \implies \lambda(E) = 0.$$

- λ is **singular** with respect to μ, written $\lambda \perp \mu$, if there are disjoint $A, B \in \mathcal{S}$ with

$$\lambda(E) = \lambda(A \cap E), \qquad \mu(E) = \mu(B \cap E), \qquad E \in \mathcal{S}. \tag{7}$$

The measure defined by (2) on the real line is absolutely continuous with respect to Lebesgue measure. The Dirac measure, on the other hand, is singular with respect to Lebesgue measure.

A linear combination of two complex measures absolutely continuous with respect to μ is again absolutely continuous. Let λ_1, λ_2 be singular with respect to μ, with disjoint $A_i, B_i \in \mathcal{S}$ such that for $i = 1, 2$

$$\lambda_i(E) = \lambda_i(A_i \cap E), \qquad \mu(E) = \mu(B_i \cap E), \qquad E \in \mathcal{S}.$$

Put $A = A_1 \cup A_2$, $B = B_1 \cap B_2$. Then A, B are disjoint and

$$(\lambda_1 + \lambda_2)(A \cap E) = (\lambda_1 + \lambda_2)(E), \qquad \mu(B \cap E) = \mu(E)$$

(indeed $\lambda_1(A \cap E) = \lambda_1(A_1 \cap A \cap E) = \lambda_1(A_1 \cap E)$, and $\mu(E) = \mu(B_1 \cap E) = \mu(B_2 \cap B_1 \cap E)$). So linear combinations of singular measures are singular.

If a complex measure λ is both absolutely continuous and singular with respect to μ, take disjoint A, B satisfying (7). For each $E \in \mathcal{S}$, $\mu(E \cap A) = \mu(E \cap A \cap B) = 0$, so $\lambda(E \cap A) = 0$ by $\lambda \ll \mu$. Hence $\lambda(E) = \lambda(E \cap A) = 0$, i.e. $\lambda = 0$.

The first part (8) of the next theorem says that, for a fixed positive measure μ on a σ-algebra, every complex measure is the sum of an absolutely continuous measure and a singular measure; this is the **Lebesgue decomposition**. The second part (9) says that the absolutely continuous part is in fact given by a function, the **Radon–Nikodym derivative** of λ with respect to μ. Together this is the Radon–Nikodym theorem, one of the most important theorems of this whole course; it makes precise the slogan "measures are generalized functions".

Theorem 7.2.1 (Lebesgue–Radon–Nikodym)

Let μ be a σ-finite positive measure and λ a complex measure on a σ-algebra $\mathcal{S}$ of X. Then there are unique measures λ_a, λ_s with

$$\lambda = \lambda_a + \lambda_s, \qquad \lambda_a \ll \mu, \qquad \lambda_s \perp \mu. \tag{8}$$

Moreover there is a unique $h \in L^1(X, \mu)$ with

$$\lambda_a(E) = \int_E h \, d\mu, \qquad E \in \mathcal{S}. \tag{9}$$

If μ and λ are σ-finite positive measures, there are unique measures λ_a, λ_s satisfying (8) and a unique measurable $h : X \to [0, \infty]$ satisfying (9).

Proof. First let λ be a complex measure; we start with uniqueness. If (λ_a, λ_s) and (λ_a', λ_s') both satisfy (8), then $\lambda_a - \lambda_a' = \lambda_s' - \lambda_s$ is both absolutely continuous and singular with respect to μ, hence 0. Uniqueness of $h \in L^1(X, \mu)$ follows from Proposition 1.2.10(b) (if $\int_E h = 0$ for all E, then $h = 0$ a.e.).

Step 1: μ and λ finite positive measures. Then $\nu = \mu + \lambda$ is also a finite positive measure. For every $f \in L^2(\nu)$, by the Cauchy–Schwarz inequality,

$$\Big| \int_X f \, d\lambda \Big| \le \int_X |f| \, d\lambda \le \int_X |f| \, d\nu \le \Big(\int_X |f|^2 \, d\nu \Big)^{1/2} \nu(X)^{1/2},$$

and since $\nu(X) < \infty$, $f \mapsto \int_X f \, d\lambda$ is a bounded linear functional on $L^2(\nu)$. By Theorem 6.3.3 (Riesz) there is $g \in L^2(\nu)$ with

$$\int_X f \, d\lambda = \int_X f g \, d\nu, \qquad f \in L^2(\nu). \tag{10}$$

As $\nu(X) < \infty$, (18) of §3.3 gives $g \in L^1(\nu)$. Moreover, putting $f = \chi_E$,

$$\nu(E) > 0 \implies \frac{1}{\nu(E)} \int_E g \, d\nu = \frac{\lambda(E)}{\nu(E)} \in [0, 1],$$

so by Proposition 3.3.5, $0 \le g \le 1$ ν-almost everywhere. Hence we may assume $0 \le g \le 1$ while keeping (10). Put

$$A = \{x \in X : 0 \le g(x) < 1\}, \qquad B = \{x \in X : g(x) = 1\},$$

and for $E \in \mathcal{S}$

$$\lambda_a(E) = \lambda(E \cap A), \qquad \lambda_s(E) = \lambda(E \cap B).$$

Rewriting (10) (using $\nu = \mu + \lambda$),

$$\int_X (1-g) f \, d\lambda = \int_X fg \, d\mu, \qquad f \in L^2(\nu). \tag{11}$$

Putting $f = \chi_B$,

$$0 = \int_B (1-g) \, d\lambda = \int_B g \, d\mu = \mu(B),$$

so $\lambda_s \perp \mu$ (with the disjoint sets B and $A = X \setminus B$: $\lambda_s(E) = \lambda_s(B \cap E)$ and $\mu(E) = \mu(A \cap E)$). Next, putting $f = (1 + g + \cdots + g^n)\chi_E$ in (11),

$$\int_E (1 - g^{n+1}) \, d\lambda = \int_E g(1 + g + \cdots + g^n) \, d\mu.$$

Let $n \to \infty$. Since $(1 - g^{n+1})|_B = 0$ and $1 - g^{n+1} \nearrow 1$ on A, the left side tends to $\lambda(E \cap A) = \lambda_a(E)$ (monotone convergence). If $h = \lim_n g(1 + g + \cdots + g^n)$, the right side tends to $\int_E h \, d\mu$ by the monotone convergence theorem. This gives (8) and (9), and in particular $\lambda_a \ll \mu$. Since $\lambda_a(X) < \infty$, $h \in L^1(\mu)$.

Step 2: μ σ-finite, λ finite positive. Write $X = \bigsqcup_n X_n$ with $\mu(X_n) < \infty$ and choose positive numbers $\langle \alpha_n \rangle$ with $\sum_n \alpha_n \mu(X_n) < \infty$. For $E \in \mathcal{S}$ put

$$\mu_0(E) = \sum_{n=1}^{\infty} \alpha_n \mu(E \cap X_n); \tag{12}$$

μ_0 is clearly a finite positive measure. By Step 1 there are λ_a, λ_s with

$$\lambda = \lambda_a + \lambda_s, \qquad \lambda_a \ll \mu_0, \qquad \lambda_s \perp \mu_0.$$

If $\mu(E) = 0$, then $\mu(E \cap X_n) = 0$ for every n, so $\mu_0(E) = 0$; hence $\lambda_a \ll \mu_0$ gives $\lambda_a \ll \mu$. If $\mu_0(E) = \mu_0(B \cap E)$, then $\mu_0(E \setminus B) = 0$, so $\mu((E \setminus B) \cap X_n) = 0$ for every n (as $\alpha_n > 0$). Hence $\mu(E \setminus B) = 0$ and $\mu(E) = \mu(E \cap B)$; so $\lambda_s \perp \mu_0$ gives $\lambda_s \perp \mu$. This proves (8). By Step 1 there is $h_0 \in L^1(\mu_0)$ with

$$\lambda_a(E) = \int_E h_0 \, d\mu_0, \qquad E \in \mathcal{S}.$$

Put

$$k = \sum_{n=1}^{\infty} \alpha_n \chi_{X_n}. \tag{13}$$

Then $\int_E d\mu_0 = \int_E k \, d\mu$, so

$$\int_X f \, d\mu_0 = \int_X fk \, d\mu$$

for every simple f, hence (by the definition of the integral) for every measurable $f : X \to [0, \infty]$. Applying this to $f = h_0 \chi_E$ and putting $h = h_0 k$ gives (9), and $h \in L^1(\mu)$ since $\int h \, d\mu = \int h_0 \, d\mu_0 < \infty$ (for $h_0 \ge 0$).

So the theorem holds when μ is a σ-finite positive measure and λ a finite positive measure. If λ is a complex measure, it is a linear combination of finite positive measures by (6), and the same conclusion follows (apply the result to each of the four and combine; uniqueness was shown above). ■

Exercise 7.2.1

Prove the Lebesgue–Radon–Nikodym theorem when λ is a σ-finite positive measure. Also prove it when λ is a signed measure that is a difference of σ-finite positive measures. Note that in the uniqueness proof one cannot subtract positive measures directly (they may be infinite).

Solution. *Existence (λ σ-finite positive).* Write $X = \bigsqcup_m Y_m$ with $\lambda(Y_m) < \infty$ and let $\lambda^{(m)}(E) = \lambda(E \cap Y_m)$, a finite positive measure. By the theorem, $\lambda^{(m)} = \lambda_a^{(m)} + \lambda_s^{(m)}$ with $d\lambda_a^{(m)} = h_m\, d\mu$, $h_m \geq 0$, and $\lambda_s^{(m)}$ concentrated on a set A_m with $\mu(A_m) = 0$. Replacing A_m by $A_m \cap Y_m$ and h_m by $h_m \chi_{Y_m}$ changes nothing (both measures vanish outside Y_m). Put

$$\lambda_a = \sum_m \lambda_a^{(m)}, \quad \lambda_s = \sum_m \lambda_s^{(m)}, \quad h = \sum_m h_m, \quad A = \bigcup_m A_m.$$

Then $\lambda = \lambda_a + \lambda_s$, $\lambda_a(E) = \int_E h\, d\mu$ (monotone convergence), $\lambda_a \ll \mu$, $\mu(A) = 0$ and $\lambda_s(E) = \lambda_s(E \cap A)$, so $\lambda_s \perp \mu$.

Uniqueness. Suppose also $\lambda = \lambda'_a + \lambda'_s$, $d\lambda'_a = h'\, d\mu$, λ'_s concentrated on A' with $\mu(A') = 0$. Let $N = A \cup A'$, so $\mu(N) = 0$. On N^c both singular parts vanish, and on N both absolutely continuous parts vanish; hence for every E

$$\lambda_s(E) = \lambda(E \cap N) = \lambda'_s(E), \qquad \lambda_a(E) = \lambda(E \setminus N) = \lambda'_a(E).$$

No subtraction was used. For h: on each $Y_m \cap X_n$ (with $\mu(X_n) < \infty$) the measure λ_a is finite, so $\int_E (h - h')\, d\mu = 0$ for all $E \subset Y_m \cap X_n$ with h, h' integrable there, giving $h = h'$ a.e. on each piece, hence a.e.

Signed case. If $\lambda = \rho_1 - \rho_2$ with ρ_i σ-finite positive measures (at least one finite, so λ is defined), decompose each $\rho_i = \rho_{i,a} + \rho_{i,s}$ as above and set $\lambda_a = \rho_{1,a} - \rho_{2,a}$, $\lambda_s = \rho_{1,s} - \rho_{2,s}$, $h = h_1 - h_2$. These are well defined (the finite ρ_i has finite parts), and linear combinations preserve $\ll \mu$ and $\perp \mu$. Uniqueness follows exactly as above using the null set N: $\lambda_s(E) = \lambda(E \cap N)$ and $\lambda_a(E) = \lambda(E \setminus N)$ for *any* decomposition. □

When λ_a and μ satisfy (9) we write

$$d\lambda_a = h\, d\mu, \qquad h = \Big[\frac{d\lambda_a}{d\mu}\Big].$$

Exercise 7.2.2

If $d\lambda = g\, d\mu$, show that $\lambda(\{x : g(x) = 0\}) = 0$.

Solution. Let $Z = \{x : g(x) = 0\}$. Then $\lambda(Z) = \int_Z g\, d\mu = \int_Z 0\, d\mu = 0$. (More generally $\lambda(E) = 0$ and even $|\lambda|(E) = 0$ for every measurable $E \subset Z$.) □

7.2.2 Polar decomposition

The Lebesgue–Radon–Nikodym theorem has many applications in measure theory; we look at a few.

Theorem 7.2.2 (polar decomposition)

For every complex measure λ on a set X there is a unique $h \in L^1(X, |\lambda|)$ with

$$d\lambda = h\, d|\lambda|, \qquad |h| = 1.$$

Proof. Since $\lambda \ll |\lambda|$, there is a unique $h \in L^1(|\lambda|)$ with $d\lambda = h\, d|\lambda|$ (Theorem 7.2.1, the singular part being 0). If $|\lambda|(E) > 0$, then

$$\Big|\frac{1}{|\lambda|(E)} \int_E h\, d|\lambda|\Big| = \frac{|\lambda(E)|}{|\lambda|(E)} \le 1,$$

so by Proposition 3.3.5, $|h| \le 1$ almost everywhere. On the other hand, for $r > 0$ put $A_r = \{x \in X : |h(x)| < r\}$. If $A_r = \bigsqcup_{n=1}^{\infty} E_n$, then

$$\sum_{n=1}^{\infty} |\lambda(E_n)| = \sum_{n=1}^{\infty} \Big| \int_{E_n} h\, d|\lambda|\Big| \le \sum_{n=1}^{\infty} r|\lambda|(E_n) = r|\lambda|(A_r),$$

so $|\lambda|(A_r) \le r|\lambda|(A_r)$, and for $r < 1$ this forces $|\lambda|(A_r) = 0$. Hence $|h| \ge 1$ almost everywhere, and we may redefine h on a null set so that $|h| = 1$. ∎

7.2.3 Integration with respect to a complex measure

The polar decomposition suggests how to define integration with respect to a complex measure. Given $d\lambda = h\, d|\lambda|$, for every $f \in L^1(X, |\lambda|)$ define

$$\int_X f\, d\lambda = \int_X fh\, d|\lambda|.$$

If we decompose λ into positive measures as in (6), then for every $f \in L^1(X, |\lambda|)$

$$\int_X f\, d\lambda = \Big(\int_X f\, d\lambda_{1+} - \int_X f\, d\lambda_{1-}\Big) + i\Big(\int_X f\, d\lambda_{2+} - \int_X f\, d\lambda_{2-}\Big). \tag{14}$$

If f is a characteristic function this is just the definition of $d\lambda = h\, d|\lambda|$, so it holds for every simple function. If $f \in L^1(X, |\lambda|)$ and $f \ge 0$, Theorem 1.2.4 gives an increasing sequence of simple functions $\langle s_n \rangle$ converging pointwise to f. Since f belongs to $L^1(X, |\lambda|)$ and to $L^1(X, \lambda_{i\pm})$ $(i = 1, 2$; note $\lambda_{i\pm} \le |\lambda|)$, the Lebesgue convergence theorem gives (14), and therefore (14) holds for every complex $f \in L^1(X, |\lambda|)$.

7.2.4 The dual of $L^p(\mu)$

Finally, another application of Radon–Nikodym: the dual of $L^p(\mu)$ is $L^q(\mu)$ $(1 \le p < \infty)$; again σ-finiteness is needed. Recall from Example 2 of §6.1 that

$$g \mapsto \varphi_g : L^q(\mu) \to L^p(\mu)^*$$

is an isometry; it remains to show it is surjective. (For $p = 1$, $q = \infty$, showing this map is an isometry used σ-finiteness of μ.) Let $\varphi \in L^p(\mu)^*$; we find $g \in L^q(\mu)$ with $\varphi = \varphi_g$.

Finite μ. For $E \in \mathcal{S}$ put

$$\lambda(E) = \varphi(\chi_E).$$

Since $\chi_{E_1 \sqcup E_2} = \chi_{E_1} + \chi_{E_2}$, $\lambda(E_1 \sqcup E_2) = \lambda(E_1) + \lambda(E_2)$. If $E = \bigsqcup_{n=1}^{\infty} E_n$ and $A_n = E_1 \sqcup \cdots \sqcup E_n$, then

$$\lim_{n\to\infty} \|\chi_E - \chi_{A_n}\|_p = \lim_{n\to\infty} \big[\mu(E \setminus A_n)\big]^{1/p} = 0$$

(here $p < \infty$ and $\mu(E) < \infty$), and since $\varphi \in L^p(\mu)^*$,

$$\lambda(E) = \lim_{n\to\infty} \lambda(A_n) = \lim_{n\to\infty} \sum_{k=1}^{n} \lambda(E_k).$$

So λ is a complex measure. Clearly $\lambda \ll \mu$ (if $\mu(E) = 0$ then $\chi_E = 0$ in L^p), so taking the Radon–Nikodym derivative $g \in L^1(\mu)$,

$$\varphi(\chi_E) = \int_E g\,d\mu = \int_X \chi_E g\,d\mu = \varphi_g(\chi_E), \qquad E \in \mathcal{S}.$$

Hence $\varphi(f) = \varphi_g(f)$ for every simple f. The simple functions are dense in $L^\infty(\mu)$, and φ, φ_g are continuous on $L^\infty(\mu)$ (for finite μ, $\|f\|_p \le \mu(X)^{1/p}\|f\|_\infty$; and $g \in L^1$), so $\varphi(f) = \varphi_g(f)$ for every $f \in L^\infty(\mu)$. It remains to show $g \in L^q(\mu)$ and $\varphi = \varphi_g$ on all of $L^p(\mu)$.

Exercise 7.2.3

For a finite positive measure μ, show that the vector space of simple functions is dense in $L^p(\mu)$ for $1 \le p \le \infty$. [Hint: treat $1 \le p < \infty$ and $p = \infty$ separately; Lemma 3.2.3.]

Solution. $p = \infty$. Let $f \in L^\infty(\mu)$; changing f on a null set, $|f| \le \|f\|_\infty = M$ everywhere. Cover the disc $\{|z| \le M\}$ by finitely many disjoint Borel sets $B_1, \dots, B_N$ of diameter $< \epsilon$, pick $z_k \in B_k$, and let $s = \sum_k z_k \chi_{f^{-1}(B_k)}$. Then $|f - s| < \epsilon$ everywhere, so $\|f - s\|_\infty \le \epsilon$. (No finiteness of μ is needed here.) $1 \le p < \infty$. By splitting into real and imaginary, positive and negative parts, assume $f \ge 0$. By Theorem 1.2.4 there are simple $0 \le s_n \nearrow f$. Then $|f - s_n|^p \le f^p \in L^1$ and $|f - s_n|^p \to 0$ pointwise, so $\|f - s_n\|_p \to 0$ by the Lebesgue convergence theorem. (Each $s_n \le f$ is in L^p; finiteness of μ ensures every simple function, including χ_X, is in L^p.) □

$p = 1$, $q = \infty$. For every $E \in \mathcal{S}$,

$$\Big|\int_E g\,d\mu\Big| = |\varphi(\chi_E)| \le \|\varphi\|\|\chi_E\|_1 = \|\varphi\|\mu(E),$$

so by Proposition 3.3.5, $\|g\|_\infty \le \|\varphi\|$.

$1 < p < \infty$. The basic method is the same as for ℓ^p. For $n = 1, 2, \dots$ put $E_n = \{x \in X : |g(x)| \le n\}$ and $f_n = \chi_{E_n}|g|^{q-1}\cdot\operatorname{sgn} g$ (with $\operatorname{sgn} g = \overline{g}/|g|$ where $g \ne 0$, so that $f_n g = \chi_{E_n}|g|^q$). Then $f_n \in L^\infty(\mu)$ and

$$\int_{E_n} |g|^q = \int_X f_n g = \varphi_g(f_n) = \varphi(f_n) \le \|\varphi\|\|f_n\|_p = \|\varphi\|\Big(\int_{E_n} |g|^q\Big)^{1/p}.$$

Dividing both sides by the last factor and applying the monotone convergence theorem,

$$\|g\|_q = \lim_{n\to\infty}\Big(\int_X \chi_{E_n}|g|^q\Big)^{1/q} \le \|\varphi\|.$$

Now that $g \in L^q(\mu)$, φ_g is continuous on $L^p(\mu)$; since simple functions are dense in $L^p(\mu)$, $\varphi(f) = \varphi_g(f)$ for every $f \in L^p(\mu)$. So $L^p(\mu)^* = L^q(\mu)$ for finite μ and $1 \le p < \infty$.

σ-finite μ. Extending to σ-finite μ is similar to the argument in the Lebesgue–Radon–Nikodym theorem. Let μ_0 and $k \in L^1(\mu)$ be defined by (12), (13), so that $d\mu_0 = k\,d\mu$. Then

$$K : f \mapsto k^{1/p} f : L^p(\mu_0) \to L^p(\mu)$$

is a surjective isometry, so $\varphi \circ K$ is a bounded linear functional on $L^p(\mu_0)$. By the finite case there is $g \in L^q(\mu_0)$ with

$$(\varphi \circ K)(f) = \int_X fg\,d\mu_0, \qquad f \in L^p(\mu_0).$$

Rewriting,

$$\varphi(Kf) = \int_X fgk\,d\mu = \int_X k^{1/p} f g k^{1/q}\,d\mu = \int_X (Kf)\cdot(gk^{1/q})\,d\mu$$

(with $1/\infty = 0$). Hence $gk^{1/q}$ is the function we were looking for.

Exercise 7.2.4

In the argument above, show that K is a surjective isometry from $L^p(\mu_0)$ onto $L^p(\mu)$, and that $gk^{1/q} \in L^q(\mu)$.

Solution. $k > 0$ everywhere (each x lies in some X_n and $\alpha_n > 0$). *Isometry:* $\int_X |k^{1/p} f|^p \, d\mu = \int_X |f|^p k \, d\mu = \int_X |f|^p \, d\mu_0$, using $\int F \, d\mu_0 = \int Fk \, d\mu$ for measurable $F \geq 0$. *Surjective:* for $u \in L^p(\mu)$, $f = k^{-1/p} u$ satisfies $\int |f|^p \, d\mu_0 = \int |u|^p k^{-1} k \, d\mu = \|u\|_p^p < \infty$ and $Kf = u$. Also μ_0 and μ have the same null sets ($k > 0$), so a.e. statements agree. $gk^{1/q} \in L^q(\mu)$*:* for $1 < q < \infty$, $\int |gk^{1/q}|^q \, d\mu = \int |g|^q k \, d\mu = \int |g|^q \, d\mu_0 < \infty$. For $q = \infty$ ($p = 1$), $k^{1/q} = 1$ and $\|g\|_{L^\infty(\mu)} = \|g\|_{L^\infty(\mu_0)} < \infty$ because the null sets agree. □

Summarizing:

Theorem 7.2.3

For a σ-finite positive measure μ on a σ-algebra of X, $L^p(\mu)^* = L^q(\mu)$, where p, q are conjugate exponents and $1 \leq p < \infty$.

Exercise 7.2.5

Let $X = \{a, b\}$ with $\mu(\{a\}) = 1$, $\mu(\{b\}) = \infty$. Is the dual of $L^1(\mu)$ equal to $L^\infty(\mu)$?

Solution. A measurable f is in $L^1(\mu)$ iff $|f(a)| + \infty \cdot |f(b)| < \infty$, i.e. iff $f(b) = 0$. So $L^1(\mu) \cong \mathbb{C}$ via $f \mapsto f(a)$, with $\|f\|_1 = |f(a)|$, and $L^1(\mu)^* \cong \mathbb{C}$. On the other hand $\{b\}$ is not a null set, so $L^\infty(\mu) \cong \mathbb{C}^2$ with the max norm. The map $g \mapsto \varphi_g$, $\varphi_g(f) = g(a)f(a)$, is surjective onto $L^1(\mu)^*$ but not injective: $g = \chi_{\{b\}}$ gives $\varphi_g = 0$ though $\|g\|_\infty = 1$. So the natural map is not an isometric isomorphism, and $L^1(\mu)^* \neq L^\infty(\mu)$; this shows σ-finiteness cannot be dropped in Theorem 7.2.3 for $p = 1$. □

Corollary 7.2.4

Let μ be a σ-finite positive measure on X and $1 \leq p \leq \infty$. If a measurable function g on X satisfies

$$h \in L^q(\mu) \implies \left| \int_X gh \, d\mu \right| \leq M \|h\|_q \qquad (p, q \text{ conjugate}),$$

then $g \in L^p(\mu)$ and $\|g\|_p \leq M$.

Proof. The hypothesis says $\varphi_g \in L^q(\mu)^*$ and $\|\varphi_g\| \leq M$. So if $1 < p \leq \infty$ (i.e. $1 \leq q < \infty$), Theorem 7.2.3 applies and the isometry gives $\|g\|_p = \|\varphi_g\| \leq M$. For $p = 1$, the hypothesis gives

$$\int_X |g| \, d\mu = \int_X g \cdot (\operatorname{sgn} g) \, d\mu \leq M \|\operatorname{sgn} g\|_\infty \leq M,$$

where $\operatorname{sgn} g = \overline{g}/|g|$ on $\{g \neq 0\}$ and 0 elsewhere. ■

Corollary 7.2.5

For every $f \in L^1(\mathbb{R})$ and $g \in L^p(\mathbb{R})$ ($1 \leq p \leq \infty$), $f * g \in L^p(\mathbb{R})$ and $\|f * g\|_p \leq \|f\|_1 \|g\|_p$.

Proof. For every $h \in L^q(\mathbb{R})$, writing $g_y(x) = g(x - y)$,

$$\begin{aligned} \left| \int_{\mathbb{R}} (f * g)(x) h(x) \, dx \right| &= \left| \int_{\mathbb{R}} \int_{\mathbb{R}} f(y) g(x - y) \, dy \, h(x) \, dx \right| \\ &\leq \int_{\mathbb{R}} \left| \int_{\mathbb{R}} g_y(x) h(x) \, dx \right| |f(y)| \, dy \\ &\leq \|g_y\|_p \|h\|_q \|f\|_1 = \|f\|_1 \|g\|_p \|h\|_q \end{aligned}$$

(Fubini, Hölder, translation invariance). Corollary 7.2.4 gives the result. ■

(To be careful, apply the computation first to $|f|, |g|, |h|$: Tonelli gives $\iint |f(y)g(x-y)h(x)|\,dy\,dx \le \|f\|_1\|g\|_p\|h\|_q < \infty$, which justifies Fubini and shows $(f * g)(x)$ is defined a.e.)

7.3 Borel Measures on the Real Line

7.3.1 Regular Borel measures and their distribution functions

In this section we find all complex measures defined on the Borel sets of the real line. As in §4.1, $\mathcal{B}$ denotes the smallest σ-algebra containing all open subsets of $\mathbb{R}$; its members are the Borel sets. A complex measure on the Borel sets is a **complex Borel measure**. Every complex measure is a linear combination of positive measures, so we focus on positive Borel measures. Here we consider positive measures whose value on every bounded interval is finite; such a Borel measure is called a **regular positive Borel measure**.

For a regular positive Borel measure λ on the real line define $\alpha_\lambda : \mathbb{R} \to \mathbb{R}$ by

$$\alpha_\lambda(x) = \begin{cases} \lambda((0,x]), & x > 0, \\ 0, & x = 0, \\ -\lambda((x,0]), & x < 0. \end{cases}$$

Then α_λ is clearly nondecreasing. It is also right-continuous. First let $x > 0$ and take any decreasing sequence $\langle h_n \rangle$ in $(0, x)$ converging to 0. Then

$$\alpha_\lambda(x + h_n) - \alpha_\lambda(x) = \lambda((0, x + h_n]) - \lambda((0, x]) = \lambda((x, x + h_n]),$$

and since $\bigcap_{n=1}^{\infty}(x, x + h_n] = \emptyset$ and $\lambda((x, x + h_1]) < \infty$, continuity from above gives

$$\lim_{n\to\infty} \{\alpha_\lambda(x + h_n) - \alpha_\lambda(x)\} = 0.$$

Thus $\lim_{h\to 0^+} \alpha_\lambda(x + h) = \alpha_\lambda(x)$. The cases $x = 0$ and $x < 0$ are handled the same way, so α_λ is right-continuous.

Exercise 7.3.1

For a regular positive Borel measure λ on the real line, show that $\alpha_\lambda : \mathbb{R} \to \mathbb{R}$ is a nondecreasing right-continuous function.

Solution. In all cases $\alpha_\lambda(b) - \alpha_\lambda(a) = \lambda((a, b])$ for $a < b$: e.g. for $a < 0 < b$, $\alpha_\lambda(b) - \alpha_\lambda(a) = \lambda((0, b]) + \lambda((a, 0]) = \lambda((a, b])$; the cases $0 \le a < b$ and $a < b \le 0$ are similar. Hence $\alpha_\lambda(b) - \alpha_\lambda(a) \ge 0$ (nondecreasing). For right-continuity at any x and $h_n \searrow 0$: $\alpha_\lambda(x + h_n) - \alpha_\lambda(x) = \lambda((x, x + h_n]) \to \lambda(\emptyset) = 0$ by continuity from above, valid since $\lambda((x, x + h_1]) < \infty$ by regularity. As α_λ is monotone, convergence along every decreasing sequence gives $\lim_{h\to 0^+} \alpha_\lambda(x + h) = \alpha_\lambda(x)$. □

7.3.2 Riemann–Stieltjes integrals

For a bounded real function $f : [a, b] \to \mathbb{R}$ consider the Riemann–Stieltjes integral $\int_a^b f\,d\alpha_\lambda$, which proceeds like the other Riemann integrals of §1.4. For a partition $P = \{x_0, x_1, \ldots, x_n\}$ of $[a, b]$, define U_P and L_P as in (23) of §1.4. Then the Riemann–Stieltjes sums are

$$\begin{aligned} U_a^b(f, P, \alpha_\lambda) &= \sum_{i=1}^{n} M_i(\alpha_\lambda(x_i) - \alpha_\lambda(x_{i-1})) = \sum_{i=1}^{n} M_i \lambda((x_{i-1}, x_i]) \\ &= \int_{[a,b]} U_P\,d\lambda - f(a)\lambda(\{a\}) = \int_{(a,b]} U_P\,d\lambda, \end{aligned}$$

and similarly $L_a^b(f,P,\alpha_\lambda) = \int_{(a,b]} L_P\, d\lambda$. Hence for every partition $P \in P[a,b]$

$$L_a^b(f,P,\alpha_\lambda) \le \int_{(a,b]} f\, d\lambda \le U_a^b(f,P,\alpha_\lambda).$$

If $f:\mathbb{R}\to\mathbb{R}$ is continuous with support in a bounded interval (A,B), then f is Riemann–Stieltjes integrable on $[A,B]$ with respect to α_λ, so

$$\int_{\mathbb{R}} f\, d\lambda = \int_A^B f(x)\, d\alpha_\lambda(x).$$

So on $C_c(\mathbb{R})$, integration with respect to a regular positive Borel measure is just Riemann–Stieltjes integration. The Riemann–Stieltjes integral is close to the general measure-theoretic integral, but its integrability becomes a big problem when the integrand is not continuous. By contrast, as remarked at the end of §1.4, integrals with respect to regular positive Borel measures or Lebesgue measure have difficulties in defining improper integrals.

7.3.3 Continuity of α_λ

We now apply the Lebesgue–Radon–Nikodym theorem to Borel measures on the real line. First we need to know when the nondecreasing function α_λ is continuous or absolutely continuous. Taking a decreasing sequence $h_n \to 0$, $\bigcap_{n=1}^\infty (x-h_n, x] = \{x\}$, so

$$\lambda(\{x\}) = \lim_{n\to\infty} \lambda((x-h_n,x]) = \lim_{n\to\infty} \{\alpha_\lambda(x) - \alpha_\lambda(x-h_n)\}.$$

So α_λ is left-continuous at x iff $\lambda(\{x\}) = 0$. A measure for which every point has measure 0 is a **continuous measure**. Since α_λ is always right-continuous:

Proposition 7.3.1

For a regular positive Borel measure λ on the real line, α_λ is continuous at x if and only if $\lambda(\{x\}) = 0$.

The next theorem explains why the term "absolutely continuous" is used for both measures and functions. To discuss absolute continuity between two measures they must be defined on the same σ-algebra; Lebesgue measure restricted to the Borel sets, $m|_{\mathcal{B}}$, is also written m.

Theorem 7.3.2

For a regular positive Borel measure λ on the real line, the following are equivalent:

(a) $\lambda \ll m$;

(b) α_λ is absolutely continuous on every bounded closed interval.

Proof. (a)⇒(b). If $\lambda \ll m$, by the Radon–Nikodym theorem there is a Borel function $h:\mathbb{R}\to[0,\infty]$ with

$$\lambda(E) = \int_E h(x)\, dx, \qquad E \in \mathcal{B}.$$

Fix a bounded closed interval I; we show α_λ is absolutely continuous on I. Since $\int_I h = \lambda(I) < \infty$, $h \in L^1(I,m)$. By Proposition 2.1.4, for every $\epsilon > 0$ there is $\delta > 0$ with

$$A \in \mathcal{B},\ m(A) < \delta \implies \lambda(A) < \epsilon.$$

This δ works in the definition of absolute continuity of functions. If $\{(a_i,b_i) : i = 1,\dots,n\}$ are disjoint intervals in I with $\sum_{i=1}^n (b_i - a_i) < \delta$, then $m(\bigcup_{i=1}^n (a_i,b_i]) < \delta$, so

$$\sum_{i=1}^n |\alpha_\lambda(b_i) - \alpha_\lambda(a_i)| = \sum_{i=1}^n \lambda((a_i,b_i]) = \lambda\Big(\bigcup_{i=1}^n (a_i,b_i]\Big) < \epsilon.$$

(b)⇒(a). Assume α_λ is absolutely continuous on every bounded closed interval I, let $\epsilon > 0$, and let $\delta_I > 0$ be the number from the definition of absolute continuity of $\alpha_\lambda : I \to \mathbb{R}$. If $m(A) = 0$ then $m(A \cap I) = 0$, so there is a sequence of disjoint open intervals $\langle (c_n, d_n) : n = 1, 2, \dots \rangle$ with

$$A \cap I \subset \bigcup_{n=1}^{\infty} (c_n, d_n), \qquad \sum_{n=1}^{\infty} (d_n - c_n) < \delta_I.$$

Writing (a_n, b_n) for the intersection of (c_n, d_n) with the interior $\operatorname{int} I$ of I,

$$A \cap \operatorname{int} I \subset \bigcup_{n=1}^{\infty} (a_n, b_n) \subset I, \qquad \sum_{n=1}^{\infty} (b_n - a_n) < \delta_I.$$

Hence $\sum_{n=1}^{N} [\alpha_\lambda(b_n) - \alpha_\lambda(a_n)] < \epsilon$ for every N, so $\sum_{n=1}^{\infty} [\alpha_\lambda(b_n) - \alpha_\lambda(a_n)] \le \epsilon$. Since α_λ is continuous (being absolutely continuous), $\alpha_\lambda(b_n) - \alpha_\lambda(a_n) = \lambda((a_n, b_n))$ (no point masses), and

$$\lambda(A \cap \operatorname{int} I) \le \lambda\Big(\bigcup_{n=1}^{\infty} (a_n, b_n) \Big) = \sum_{n=1}^{\infty} \lambda((a_n, b_n)) = \sum_{n=1}^{\infty} [\alpha_\lambda(b_n) - \alpha_\lambda(a_n)] \le \epsilon.$$

So $\lambda(A \cap \operatorname{int} I) = 0$ for every bounded open interval $\operatorname{int} I$, and therefore $\lambda(A) = 0$. ■

7.3.4 Regularity

We now look for the condition on α_λ for $\lambda \perp m$. First some properties of regular positive Borel measures on the real line.

Theorem 7.3.3 (regularity)

A regular positive Borel measure λ on the real line satisfies, for every Borel set $E \in \mathcal{B}$,

$$\lambda(E) = \inf\{\lambda(U) : U \supset E,\ U \text{ open}\},$$
$$\lambda(E) = \sup\{\lambda(K) : K \subset E,\ K \text{ compact}\}.$$

Proof. Clearly every open set satisfies the first identity. Since every open set is the union of an increasing sequence of compact sets, the second identity also holds for open E. Let $\mathcal{C}$ be the collection of Borel sets satisfying both identities; it suffices to show $\mathcal{C}$ is a σ-algebra.

Countable unions. Take $\langle E_n \rangle$ in $\mathcal{C}$ and $E = \bigcup_n E_n$. First the case $\lambda(E) < \infty$: choose open $U_n \supset E_n$ with $\lambda(U_n) < \lambda(E_n) + \epsilon/2^n$ and let U be their union. Since $U \setminus E \subset \bigcup_n (U_n \setminus E_n)$,

$$\lambda(U) - \lambda(E) = \lambda(U \setminus E) \le \sum_{n=1}^{\infty} \lambda(U_n \setminus E_n) < \epsilon,$$

so E satisfies the first identity. If $\lambda(E) = \infty$ the first identity is trivial. For the second, for each j choose compact sets $\langle K_{ij} : i = 1, 2, \dots \rangle$ with

$$K_{ij} \subset E_j, \qquad \lim_{i \to \infty} \lambda(K_{ij}) = \lambda(E_j).$$

With $K_n = \bigcup_{i,j=1}^{n} K_{ij}$ we can show $\lambda(K_n) \to \lambda(E)$, so $E \in \mathcal{C}$ (Exercise 7.3.2).

Complements. Let $E \in \mathcal{C}$ and choose open $U_n \supset E$ decreasing with $\lambda(U_n) \to \lambda(E)$. With $L_n = [-n, n] \setminus U_n$ (compact), $L_n \subset E^c$ and $\lambda(L_n) \to \lambda(E^c)$ (Exercise 7.3.2). It remains to show the first identity for E^c; for this we first show

$$I \text{ a bounded closed interval, } I \cap E \in \mathcal{C} \implies I \setminus E \in \mathcal{C}.$$

Choose an open $U \supset I \cap E$ with $\lambda(U) < \lambda(I \cap E) + \epsilon$. Then $K = I \setminus U$ is compact, $K \subset I \setminus E$, and since $\lambda(I) < \infty$,

$$\lambda(K) = \lambda(I) - \lambda(I \cap U) \geq \lambda(I) - \lambda(U) > \lambda(I) - \lambda(I \cap E) - \epsilon = \lambda(I \setminus E) - \epsilon,$$

so $I \setminus E$ satisfies the second identity. For the first, choose a bounded open interval $J \supset I$ with $\lambda(J) < \lambda(I) + \epsilon$, and a compact $K \subset I \cap E$ with $\lambda(K) > \lambda(I \cap E) - \epsilon$. Put $U = J \setminus K$; then U is open, $U \supset I \setminus E$, and

$$\lambda(U) = \lambda(J) - \lambda(K) < \lambda(I) + \epsilon - \lambda(I \cap E) + \epsilon = \lambda(I \setminus E) + 2\epsilon,$$

so $I \setminus E$ satisfies the first identity.

If $E \in \mathcal{C}$, then $E \cap [-n, n] \in \mathcal{C}$ for each n (one checks this as above), so $[-n, n] \setminus E \in \mathcal{C}$. Since $\mathcal{C}$ is already closed under countable unions, $E^c = \bigcup_n ([-n, n] \setminus E) \in \mathcal{C}$. ■

Exercise 7.3.2

Show that every open set is the union of an increasing sequence of compact sets. In the proof of Theorem 7.3.3 show that $\lim_n \lambda(K_n) = \lambda(E)$ and $\lim_n \lambda(L_n) = \lambda(E^c)$.

Solution. *Open sets.* For open $U \subset \mathbb{R}$ ($U \neq \mathbb{R}$) let $C_n = \{x \in U : |x| \leq n,\ d(x, U^c) \geq \frac{1}{n}\}$; for $U = \mathbb{R}$ let $C_n = [-n, n]$. Each C_n is closed and bounded, hence compact, $C_n \subset C_{n+1}$, and every $x \in U$ has $d(x, U^c) > 0$, so $x \in C_n$ for large n.

$\lambda(K_n) \to \lambda(E)$. $K_n = \bigcup_{i,j \leq n} K_{ij}$ is compact, $K_n \subset K_{n+1}$ and $K_n \subset E$, so $L = \lim_n \lambda(K_n) \leq \lambda(E)$ exists. If some $\lambda(E_j) = \infty$, then $L \geq \lambda(K_{ij}) \to \infty$ as $i \to \infty$, so $L = \lambda(E) = \infty$. Otherwise fix N and i. For $n \geq \max(i, N)$, $K_n \supset \bigcup_{j \leq N} K_{ij}$, and

$$\lambda\Big(\bigcup_{j \leq N} E_j\Big) - \lambda\Big(\bigcup_{j \leq N} K_{ij}\Big) \leq \sum_{j \leq N} \lambda(E_j \setminus K_{ij}) = \sum_{j \leq N} (\lambda(E_j) - \lambda(K_{ij})) \xrightarrow[i \to \infty]{} 0.$$

Hence $L \geq \lambda(E_1 \cup \cdots \cup E_N)$ for every N, and continuity from below gives $L \geq \lambda(E)$.

$\lambda(L_n) \to \lambda(E^c)$. Suppose $\lambda(E) < \infty$. $L_n = [-n, n] \setminus U_n$ is compact, increasing (as U_n decreases), and $L_n \subset E^c$. Since $([-n, n] \cap E^c) \setminus L_n = [-n, n] \cap U_n \cap E^c \subset U_n \setminus E$,

$$0 \leq \lambda([-n, n] \cap E^c) - \lambda(L_n) \leq \lambda(U_n \setminus E) = \lambda(U_n) - \lambda(E) \to 0,$$

while $\lambda([-n, n] \cap E^c) \to \lambda(E^c)$ by continuity from below. Hence $\lambda(L_n) \to \lambda(E^c)$. (If $\lambda(E) = \infty$, apply this to the sets $E \cap [-N, N]$, which have finite measure; this is exactly how the end of the proof reduces to $[-n, n] \setminus E$.) □

Corollary 7.3.4

If $\lambda \ll m$ and $\alpha'_\lambda = 0$ almost everywhere, then $\lambda = 0$.

Proof. By Theorem 7.3.2, α_λ is absolutely continuous, so Theorem 2.2.4 gives

$$\lambda((a, b)) = \alpha_\lambda(b) - \alpha_\lambda(a) = \int_a^b \alpha'_\lambda(x)\, dx = 0.$$

Hence $\lambda(U) = 0$ for every open U (a countable disjoint union of open intervals), and Theorem 7.3.3 gives $\lambda = 0$. ■

7.3.5 Singular measures

Theorem 7.3.5

For a regular positive Borel measure λ on the real line, the following are equivalent:

(a) $\lambda \perp m$;

(b) $\alpha'_\lambda = 0$ almost everywhere.

Proof. (b)⇒(a). Assume $\alpha'_\lambda = 0$ a.e. Write the Lebesgue decomposition $\lambda = \lambda_a + \lambda_s$; then $\alpha_\lambda = \alpha_{\lambda_a} + \alpha_{\lambda_s}$, both nondecreasing, so $0 \le \alpha'_{\lambda_a} \le \alpha'_\lambda = 0$ and hence $\alpha'_{\lambda_a} = 0$ almost everywhere. Since $\lambda_a \ll m$, Corollary 7.3.4 gives $\lambda_a = 0$, so $\lambda = \lambda_s$ is singular.

(a)⇒(b). Assume $\lambda \perp m$ and put

$$\nu(E) = \int_E \alpha'_\lambda \, dm, \qquad E \in \mathcal{B}. \tag{15}$$

Then obviously $\nu \ll m$. On the other hand, by Proposition 2.1.3,

$$\nu((a,b]) = \int_{(a,b]} \alpha'_\lambda \, dm \le \alpha_\lambda(b) - \alpha_\lambda(a) = \lambda((a,b]).$$

Every open set is a countable disjoint union of intervals of the form $(a,b]$, so $\nu(U) \le \lambda(U)$ for every open U. Since ν and λ are both regular positive Borel measures, Theorem 7.3.3 gives $\nu(E) \le \lambda(E)$ for every Borel E, hence $\nu \ll \lambda$. Since also $\lambda \perp m$, we get $\nu \perp m$ (Exercise 7.3.3), and together with $\nu \ll m$ this forces $\nu = 0$. As $\alpha'_\lambda \ge 0$ almost everywhere, $\alpha'_\lambda = 0$ almost everywhere. ■

Exercise 7.3.3

Show that if $\nu \ll \lambda$ and $\lambda \perp m$, then $\nu \perp m$.

Solution. Choose disjoint A, B with $\lambda(E) = \lambda(E \cap A)$ and $m(E) = m(E \cap B)$ for all E. Then $\lambda(E \setminus A) = 0$, so $\nu(E \setminus A) = 0$ by $\nu \ll \lambda$, i.e. $\nu(E) = \nu(E \cap A)$. Together with $m(E) = m(E \cap B)$, the disjoint pair A, B shows $\nu \perp m$. □

7.3.6 The Radon–Nikodym derivative is α'_λ

For an arbitrary regular positive Borel measure λ with Lebesgue decomposition $\lambda = \lambda_a + \lambda_s$, define ν by (15). By Theorem 7.3.5,

$$\alpha'_\lambda = \alpha'_{\lambda_a} + \alpha'_{\lambda_s} = \alpha'_{\lambda_a}$$

almost everywhere, and by Theorem 2.2.4 (α_{λ_a} absolutely continuous by Theorem 7.3.2)

$$\nu((c,d)) = \int_{(c,d)} \alpha'_\lambda \, dm = \int_{(c,d)} \alpha'_{\lambda_a} \, dm = \alpha_{\lambda_a}(d) - \alpha_{\lambda_a}(c) = \lambda_a((c,d)).$$

Hence $\nu(U) = \lambda_a(U)$ for every open U, and since ν is also regular, $\nu = \lambda_a$ (Theorem 7.3.3). That is,

$$\lambda_a(E) = \int_E \alpha'_\lambda \, dm$$

for every Borel set $E \in \mathcal{B}$: $\alpha'_{\lambda_a} = \alpha'_\lambda$ is precisely the Radon–Nikodym derivative of λ_a with respect to m.

Corollary 7.3.6

If a regular positive Borel measure λ on the real line is absolutely continuous with respect to Lebesgue measure m, then $\left[\dfrac{d\lambda}{dm}\right] = \alpha'_\lambda$ almost everywhere.

7.3.7 Discrete, continuous singular, and absolutely continuous parts

For a regular positive Borel measure λ put

$$D = \{x \in \mathbb{R} : \lambda(\{x\}) > 0\},$$

and for $E \in \mathcal{B}$

$$\lambda_d(E) = \lambda(E \cap D), \qquad \lambda_c(E) = \lambda(E \setminus D).$$

For each n, $\lambda(D \cap [-n,n]) \le \lambda([-n,n]) < \infty$, so $D \cap [-n,n]$ is countable (for each k, at most $k\lambda([-n,n])$ points have mass $> 1/k$); hence D is countable, $D = \{x_n : n = 1,2,\dots\}$. Then for every $E \in \mathcal{B}$

$$\lambda_d(E) = \sum_{x_n \in E} \lambda(\{x_n\}) = \sum_{n=1}^{\infty} \lambda(\{x_n\})\delta_{x_n}(E),$$

so λ_d is a sum of Dirac measures. The resulting nondecreasing function α_{λ_d} has a discontinuity at each point $x \in D$. On the other hand, by Proposition 7.3.1 the nondecreasing function α_{λ_c} is continuous. If $\lambda_c \perp m$, then by Theorem 7.3.5 its derivative is 0 almost everywhere. A typical example of a non-constant continuous function whose derivative vanishes almost everywhere is the Cantor function.

Summary

Every regular positive Borel measure λ on the real line is the sum of three measures:

$$\lambda = \lambda_a + \lambda_s + \lambda_d,$$

an absolutely continuous measure λ_a, a continuous singular measure λ_s, and a discrete (Dirac) measure λ_d. With the corresponding nondecreasing functions $\alpha_a, \alpha_s, \alpha_d$: α_a is absolutely continuous, α_s is a continuous function whose derivative is 0 almost everywhere, and α_d is a discontinuous (jump) function. If α is the nondecreasing function of λ, then $\alpha = \alpha_a + \alpha_s + \alpha_d$ and

$$\lambda_a(E) = \int_E \alpha_a' \, dm = \int_E \alpha' \, dm, \qquad E \in \mathcal{B}.$$

Given a regular positive Borel measure λ, there is a corresponding nondecreasing right-continuous function α, and the two linear functionals on $C_c(\mathbb{R})$

$$f \mapsto \int_{\mathbb{R}} f \, d\lambda, \qquad f \mapsto \int_{\operatorname{supp} f} f(x) \, d\alpha(x)$$

coincide; the left is the integral with respect to the Borel measure λ, the right the Riemann–Stieltjes integral with respect to the nondecreasing function α. In the next chapter we shall see that, conversely, every nondecreasing function gives rise to a corresponding regular positive Borel measure.